Calculus II Assignment 1

1. Rewrite the following series using the notation $\sum$; determine convergence or divergence of the series, and if the series converges, evaluate its sum.

   (a) $\ln\frac{2}{1}+\ln\frac{3}{2}+\ln\frac{4}{3}+\ln\frac{5}{4}+\cdots$;

   (b) $\frac{2}{3}+\frac{2}{3^2}+\frac{2}{3^3}+\frac{2}{3^4}+\cdots$;

   (c) $\left(\frac{1}{2}+\frac{1}{3}\right)+\left(\frac{1}{2^2}+\frac{1}{3^2}\right)+\left(\frac{1}{2^3}+\frac{1}{3^3}\right)+\cdots$;

   (d) $\frac{1}{2}+\frac{3}{4}+\frac{5}{6}+\frac{7}{8}+\cdots$.

2. Determine convergence or divergence of the series.

   (a) $\sum_{n=1}^{\infty}(\sqrt{n+1}-\sqrt{n})$;

   (b) $\frac{1}{1\cdot 3}+\frac{1}{3\cdot 5}+\frac{1}{5\cdot 7}+\cdots+\frac{1}{(2n-1)(2n+1)}+\cdots$;

   (c) $\sin\frac{\pi}{6}+\sin\frac{2\pi}{6}+\cdots+\sin\frac{n\pi}{6}+\cdots$.

3. Give another example of a divergent series $\sum_{n=1}^{\infty}a_n$ with $a_n\to 0$ as $n\to\infty$.

4. Evaluate the sum of the following series:

$$\frac{1}{1\cdot 2\cdot 3}+\frac{1}{2\cdot 3\cdot 4}+\cdots+\frac{1}{n\cdot(n+1)\cdot(n+2)}+\cdots$$

   [Hint: find constants a, b, c such that

$$\frac{1}{n\cdot(n+1)\cdot(n+2)}=\frac{a}{n}+\frac{b}{n+1}+\frac{c}{n+2}.]$$

5. Determine convergence or divergence of the series.

   (a) $-\frac{8}{9}+\frac{8^2}{9^2}-\frac{8^3}{9^3}+\cdots+(-1)^n\frac{8^n}{9^n}+\cdots$;

   (b) $\frac{1}{3}+\frac{1}{6}+\frac{1}{9}+\cdots+\frac{1}{3n}+\cdots$;

   (c) $\frac{1}{3}+\frac{1}{\sqrt{3}}+\frac{1}{\sqrt[3]{3}}+\cdots+\frac{1}{\sqrt[n]{3}}+\cdots$;

   (d) $\frac{3}{2}+\frac{3^2}{2^2}+\frac{3^3}{2^3}+\cdots+\frac{3^n}{2^n}+\cdots$.

6. Determine convergence or divergence of the series.

   (a) $\frac{1}{5+1^2}+\frac{1}{5+2^2}+\frac{1}{5+3^2}+\cdots$;

   (b) $\sum_{n=1}^{\infty}\frac{1}{\sqrt{n(n^4+1)}}$;

   (c) $\sum_{n=1}^{\infty}\frac{1}{1+a^n}\quad(a>0)$;

   (d) $\sum_{n=1}^{\infty}\frac{1}{\ln(1+n)}$;

   (e) $\sum_{n=1}^{\infty}\sin\frac{1}{n}$;

(f) $1 + r\cos^2\theta + r^2\cos^2 2\theta + r^3\cos^2 3\theta + \cdots \quad (0 < r < 1)$;

(g) $1 + \frac{1}{3} + \frac{1}{5} + \frac{1}{7} + \frac{1}{9} + \cdots$;

(h) $\sum_{n=1}^{\infty}\left(1 - \cos\frac{x}{n}\right)$;

(i) $\sum_{n=1}^{\infty}\ln\left(1 + \frac{x}{n}\right) \quad (x > 0)$.